

Team 86 Project Technical Report for the IREC 2021

Design and Construction of Solid Experimental Sounding Rocket, EFB (Explosive Flying Burrito)

Palestino-Díaz I.¹, Paredes-Espinosa D.²

Escuela de Ingeniería y Ciencias, Instituto Tecnológico y de Estudios Superiores de Monterrey,

Navarro-Gopfert I.A.³, Treviño-Rodríguez J.L.⁴, Vacas-Moncayo S.N.⁵, Treviño-Gómez H.⁶, González-Daccarett
M.⁷, García-Carrasco A.⁸

Av. Eugenio Garza Sada 2501 Col. Tecnológico, Monterrey, N.L., México 64700

Ortiz-Polo M., Fernández-Echeverría Y., Ruiz-Hernández P.C., Barrera-Salazar J., Carmona-Gallegos S.Z.,
Vilchis-Lagunes N., Gutiérrez-Hernández A.

Av. Eugenio Garza Sada 2501 Col. Tecnológico, Monterrey, N.L., México 64700

Abstract

EXPLOSIVE FLYING BURRITO (EFB) is a solid propulsion rocket designed to compete at the SA Cup 2021 in the 10 000 ft category. EFB is designed to carry a state of the art nanotechnology based sensor capable of detecting atmospheric pollutants with higher sensibility and lifetime compared to traditional sensors.

Each subsystem will validate the requirements outlined by the SAC's rules and requirements documentation.

The biggest challenge we faced during the competition was being in a pandemic and having team members in different states, even at different times. Communication became more complicated but we did overcome every challenge that was presented. We are proud that in our first year in the competition we achieved very satisfactory results and we managed to design a functional experimental rocket with an objective mission.

¹Project Manager, Space Makers Rockets, and Address/Mail Stop for first author.

²Engineering Lead, Space Makers Rockets, and Address/Mail Stop for second author.

³Payload Lead, Space Makers Rockets, and Address/Mail Stop for third author.

⁴Logistics Lead, Space Makers Rockets, and Address/Mail Stop for third author.

⁵Propulsion Lead, Space Makers Rockets, and Address/Mail Stop for fourth author.

⁶Recovery Lead, Space Makers Rockets, and Address/Mail Stop for fourth author.

⁷Aerostructures Lead, Space Makers Rockets, and Address/Mail Stop for fourth author.

⁸Aerostructures Co-Lead, Space Makers Rockets, and Address/Mail Stop for fourth author.

I. Introduction

Space Maker is a multidisciplinary aerospace research group composed by passionate undergraduate students and professionals at Tecnológico de Monterrey. Space Maker has participated in national and international competitions aimed for the development of space-graded missions *i.e.* United Nations/Airbus Technical Assistance Programme on Bartolomeo – All in one space mission service, Congreso Nacional de Actividades Espaciales de la Agencia Espacial Mexicana with success.

Founded in 2020, the rocket laboratory mission is to build and develop rockets in order to establish the bases for rocket engineering in Mexico. With 36 members from across the country, the Rocket Laboratory began its operations with the design of the Explosive Flying Burrito (EFB) in order to compete in the Spaceport America Cup 2021.

EFB project team is divided in 6 areas: aerostructures, avionics, payload, propulsion, recovery and project management. The latter includes logistics, legal affairs and business.

Figure 1 shows the EFB team organigram. Each area has a leader which is in charge of setting the milestones for their area and assigning the tasks.

All the areas are in constant communication with the rest of the team in order to fulfill the IREC requirements without compromising the vehicle's integrity nor the experiment necessities (See Section III Payload)

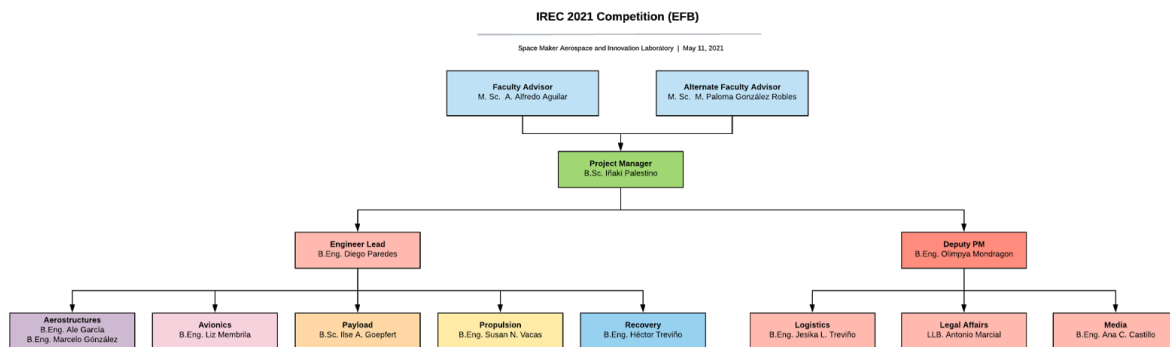


Figure 1: EFB Team Organigram

II. Flight mission (CONOPS)

The Explosive Flying Burrito Mission consists in the development of a solid propulsion rocket capable of reaching 10 000 ft in order to measure air pollutants such as H₂ and CH₄ with nanotechnology-based sensors and the correct deployment of our first air brake system.

Prior to the launch, the vehicle is rigorously inspected for any structural damage and avionics subsystem correct functioning. The latter includes the evaluation of the signal quality and the correct connection with payload.

Once the vehicle has been checked and approved to flight, the Logistic leader coordinates the launch team in order to lay the rocket at the rail launch.

Once the rocket has been positioned at the rail the pre-launch process will start. We must verify all systems are ready and secure to initiate the flight.

The launch will be started at the signal of the end of the countdown to the “fire” signal, at that moment the propulsion system it's going to prepare the chamber pressure. During the ascent the burning motor will give impulse to the rocket until the propellants are completely burned and are exhausted until they reach the apogee.

Before apogee the air brakes system is released to increase drag of the EFB to be able to reach a better precision at the apogee. at the appropriate height the NTBS register the data obtained at the mission. during all flight duration the data will be collected and monitored.

Once the descent is carried out the recovery system will open the first parachute to control the fall of the rocket, then 120 seconds later the second parachute will open to slow down the rocket's to control descent to the ground until the rocket landing will be located by the data of length, speed, acceleration and location send to the GPS system for the recovery of the rocket.

The architecture of the mission is shown in Figure 2 while Table 1 depicts the stages of the mission.

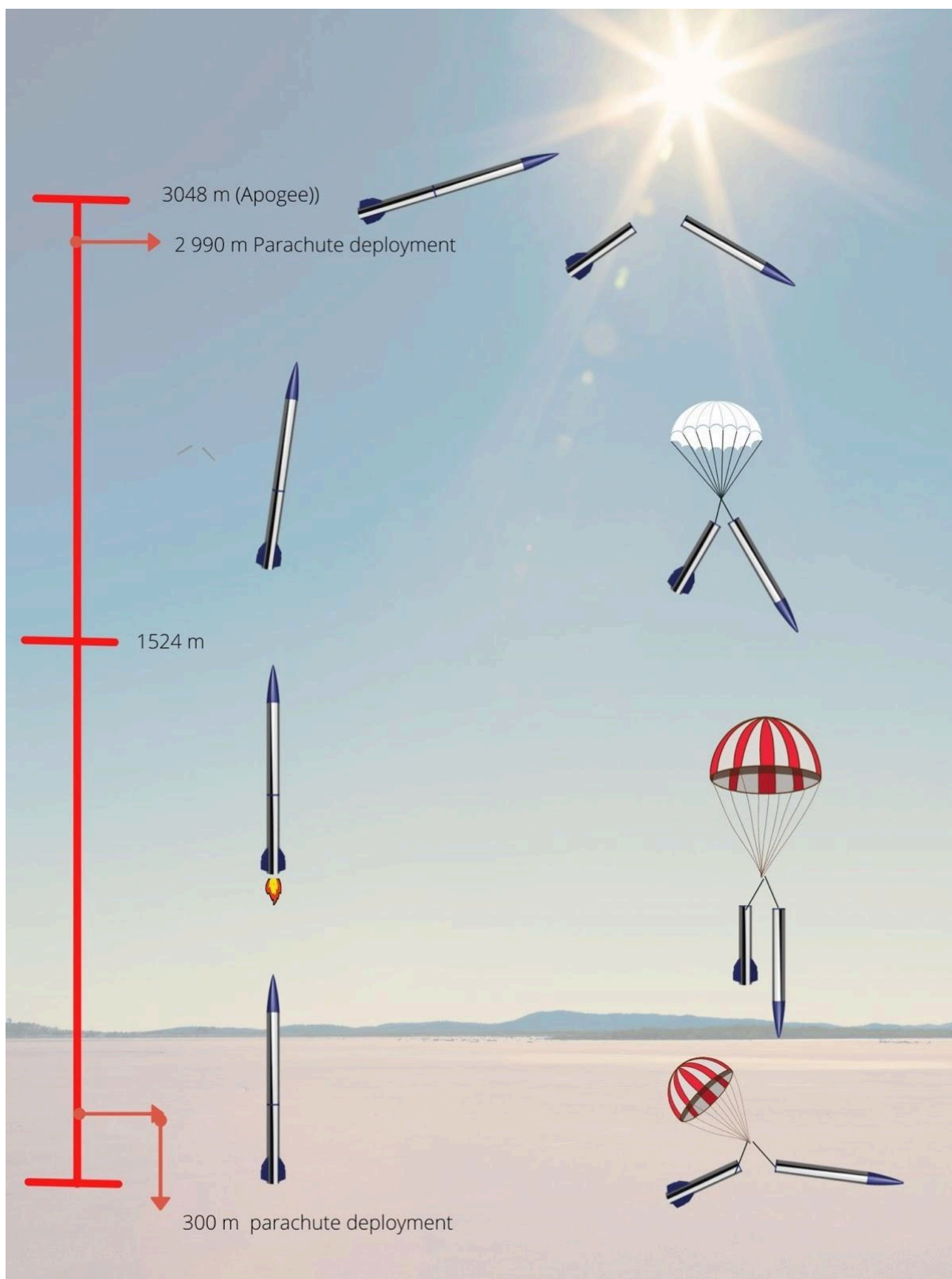


Figure 2: EFB Mission Architecture diagram

Table 1: EFB Mission Stages

Stage	Event	Description	Time (s)
1	Pre-launch	Inspection of all systems and arming protocol	N/A
2	Explosive Flying Burrito Launch	A countdown is made and the ignition occurs, soon after the rocket lifts off	0.00
3	Powered flight	The rocket is accelerated by the motor thrust	0.01 - 3.56
4	Unpowered flight	After motor burnout the rocket coasts while losing velocity	3.57 - 8.57
5	Airbrake System Deployment	Air breaks deploy and reduce rocket velocity to reach the desired altitude precisely	8.58 - 25.6
6	Apogee	Vehicle reaches nominal altitude	25.6
7	NTBS data gathering	Payload sensors gather atmospheric data in microgravity conditions	25.6
8	Drogue chute deployment	Drogue chute is deployed and descent velocity drops to 26m/s	30.6
9	Main chute deployment	Main chute is deployed and descent velocity drops to 6.4m/s	154.3
10	EFB Landing	The COTS and SRAD GPS systems will help locate the rocket	198.26

Pre-launch

Firstly, we must verify that the protocol is ready to launch the rocket by checking all the safety parameters and assuring they are all checked to initiate a safe flight and that all components are armed and integrated properly.

A. Ascent**a. Lift-off**

By this moment, we can run the launching protocol by activating the igniters with a signal in order to power on the motor safely and start the lift-off of the rocket.

b. Thrust

In this stage the rocket is being accelerated as a consequence of the KNSB burning. During this stage, the motor is being burned out.

c. Deceleration

When the motor has burned out, this stage begins. The rocket will still be getting higher while losing acceleration as it reaches its maximum altitude. This altitude must be measured by the electronic system expecting to reach 10,000 ft.

B. Airbrake Deployment

5 seconds after burnout the airbrake system will commence operation, reducing the rocket velocity to precisely the one needed to reach 10,000ft AGL

C. NTBS Data Gathering

These sensors are in charge of measuring atmospheric pollutants using sensors with nanotechnology in order to contribute to future studies for our atmosphere.

a. Real-time flight information transmission

During all flight duration, different data must be collected and monitored in real-time.

D. Descent

a. Drogue parachute

5 seconds after reaching the apogee, the 3 ft diameter drogue parachute will deploy as a phase transition. This drogue parachute is designed to assure a fast and controlled descent.

b. Main parachute

When the rocket is just below 1000 ft altitude, the 10 ft diameter main parachute is expected to deploy with a significantly slower rate of descent than the drogue one in order to have a slow and controlled descent and prepare for a safe and successful landing.

E. Landing

Thanks to the descent protocol, the rocket is now enormously decelerated from free fall, allowing a safe landing and a soft touch to the ground. Landed rocket can be found via the integrated GPS.

III. Arming sequence

The EFB Arming Sequence will be made in 16 steps prior to launchment. These steps ensure the integrity of the rocket and the payload once the vehicle has started the mission. The arming sequence protocol is shown in Table 2.

Table 2: EFB Arming sequence protocol

Step	Stage
1	All non-essential loading personnel leave
2	Payload telemetry check
3	Attachment of the payload to the vehicle
4	Closing and bolting of the hatch
5	All non-essential arming personnel leave
6	Remove the nozzle plug
7	Install igniter
8	Final visual check
9	All personnel leave the launch pad
10	Connect battery to the launch box
11	Check for no current flow through the launch box output
12	Connect igniter leads to the launch box
13	Run a final overall system check
14	Insert both launch keys
15	Turn both launch keys
16	Await for launch command

IV. Experiment (Payload description)

The Explosive Flying Burrito payload is a 3 unit cubesat. EFB's payload consists of nanotechnology-based sensors composed of Single Walled Carbon Nanotubes with an aerogel and polymer reinforcement in order to measure chemical species abundance at a certain altitude.

Carbon nanotubes were selected as the material for creating the sensors due to their amazing properties. These provide a high energy rate absorption thanks to their hollow structures, a high electrochemical accessibility on its porous surface, an impressive electrical conductivity and useful mechanical properties, making them incredible real time detectors. When interacting with compounds, carbon nanotube sensors provide immediate response to what was detected through a direct electron transfer, determining the chemical reagents of the medium.

The Payload and the air brakes will share a compartment in order to reach the conditions needed for the experiment. The payload requires interaction with the medium to test its detection properties under these microgravity conditions, so it needs to have a short entrance of airflow. The same plate that grips the continuous rotation servo of the cover is going to be used for the servo of the air brakes.

In order to validate our proposal, the payload will be exposed to the environment once the rocket begins its uncontrolled descent phase. Exposure of the payload and microgravity conditions will be achieved thanks to the open areas left from the airbrake system deployment and the vehicle's descent.

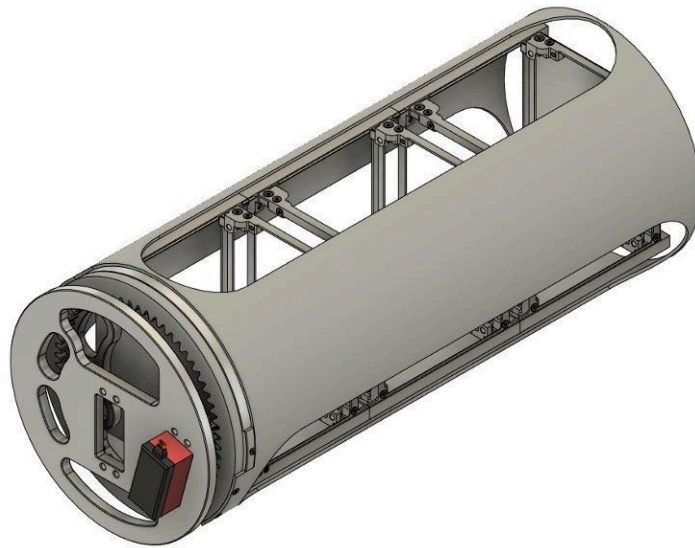


Figure 3: Payload module

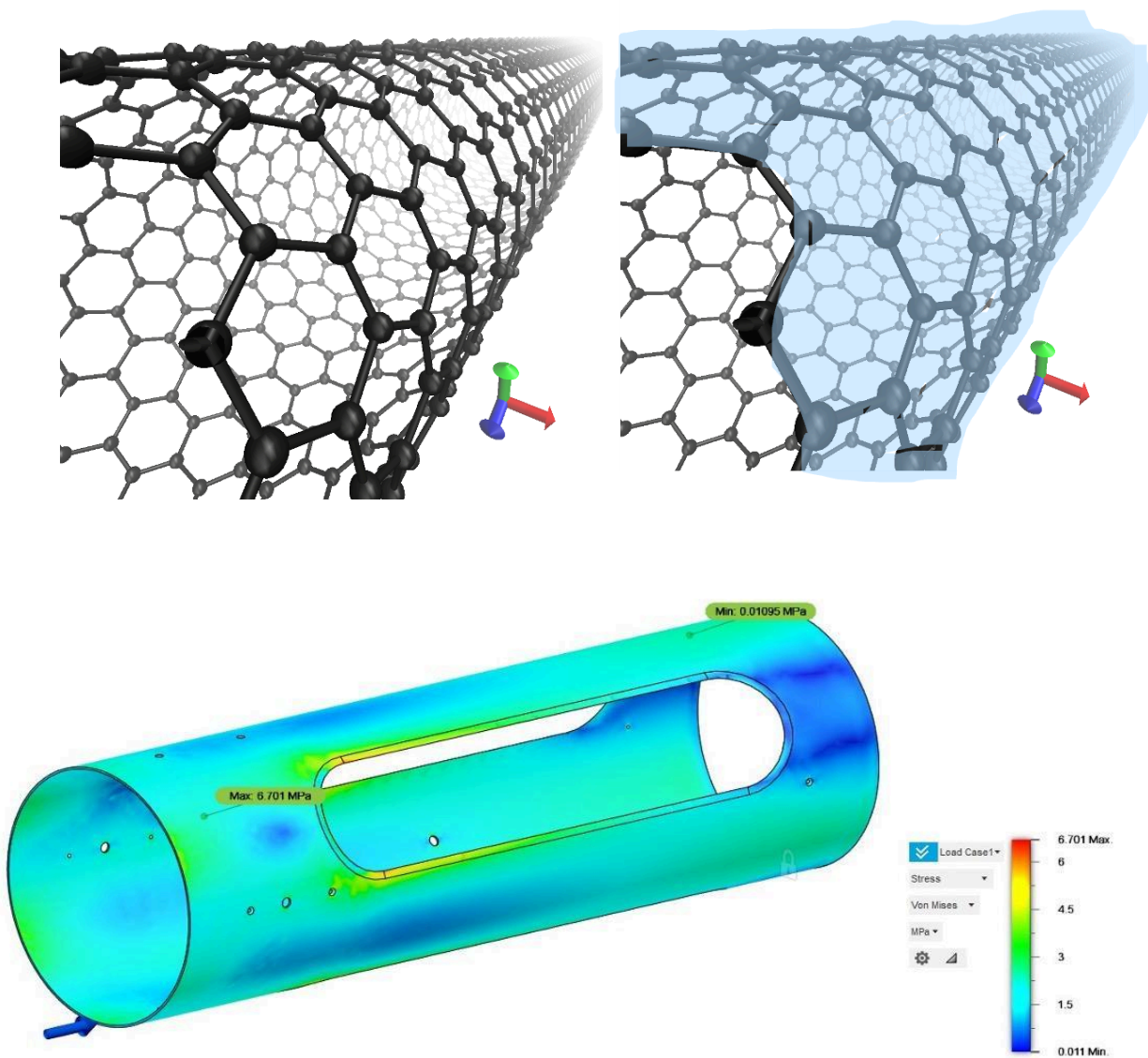


Figure 4: Upper left: Single Walled Carbon Nanotube. b) Upper Right Single Walled Carbon Nanotube with the polymer and aerogel reinforcement. c) Bottom

V. Systems Overview

The development of the EFB is a sum of efforts from our subsystems that focus on aerostructures, avionics, propulsion, payload and recovery in the Space Maker Rocket Laboratory.

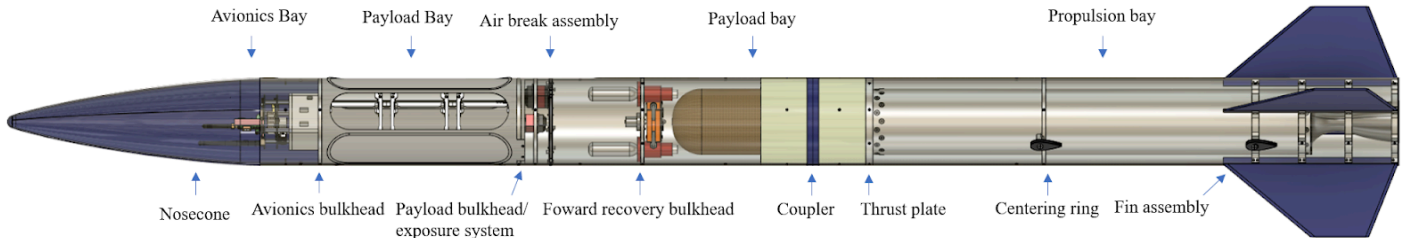


Figure 5: EFB structure

1. Aerostructures

General aspects of the rocket's airframe

The launch vehicle has a length of 2400 mm, a 152.4 mm outer diameter, and a 149.91 mm inner diameter. The forward body tube houses the avionics, payload and forward recovery bay, whereas the lower body tube houses the lower recovery bay, the booster bay and fin assembly. Both sections are made of 6061-T6 seamless aluminum tubing, whereas the nose cone is made of fiberglass due to its RF transparency.

Simulations were made using SolidWorks, Fusion 360 and OpenRocket with the purpose of ensuring the structural integrity of the rocket, while making sure IREC regulations such as the off-the-rail velocity and stability margins were met.

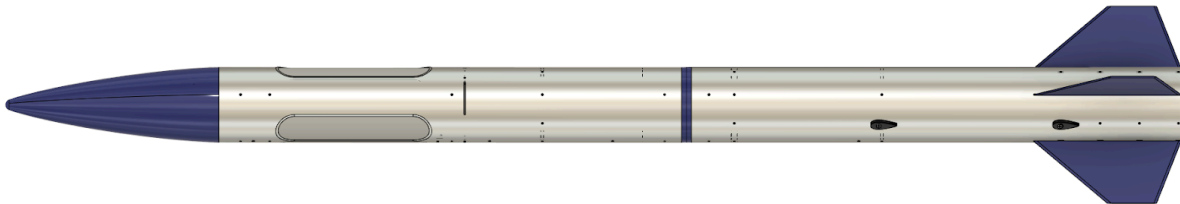


Figure 6: Basic EFB outer structure

Each component of the launch vehicle will be described in detail, following a top-down approach, and will be divided in 4 main sections: the nose cone, forward and lower body tube and the fin assembly.

A. Nosecone

An LD Haack - shaped nose cone was chosen for EFB due to its performance in transonic velocities, it is made of fiberglass because of its lack of interference for transmitting radio frequency signals. The latter is necessary for the avionics subsystem proper functioning (Refer to Avionics). It is, 434 mm long and has a maximum diameter of 152.4mm, just as the main bodytube of the launch vehicle, and a fineness ratio of 3, it is held to the body tube using four M2 flathead screws and both aerodynamic and stress tests were carried out in SolidWorks to ensure its efficiency and structural safety.

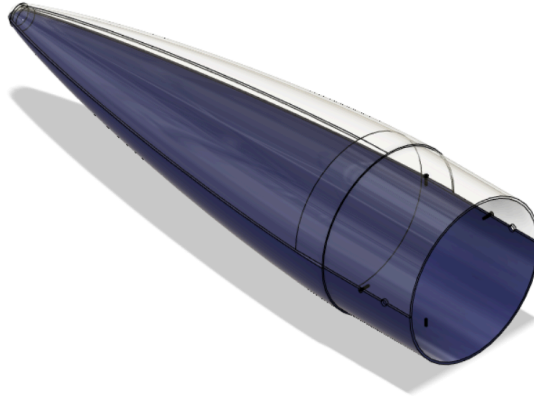


Figure 7: Nose cone blueprints

B. Upper body tube

The upper body tube is 950 mm long, it houses the nose cone, payload, air brakes and forward recovery bays. The avionics system sits on the very top of the body tube to enable RF transparency. Below it the 8.8 lb payload is held using the avionics and payload bulkhead, this section of the body tube has three large slots that allow airflow to the payload at apogee. Prior to apogee these slots are covered by the payload exposure mechanism, a concentric cylinder with matching slots that rotates to open up airflow to the payload.

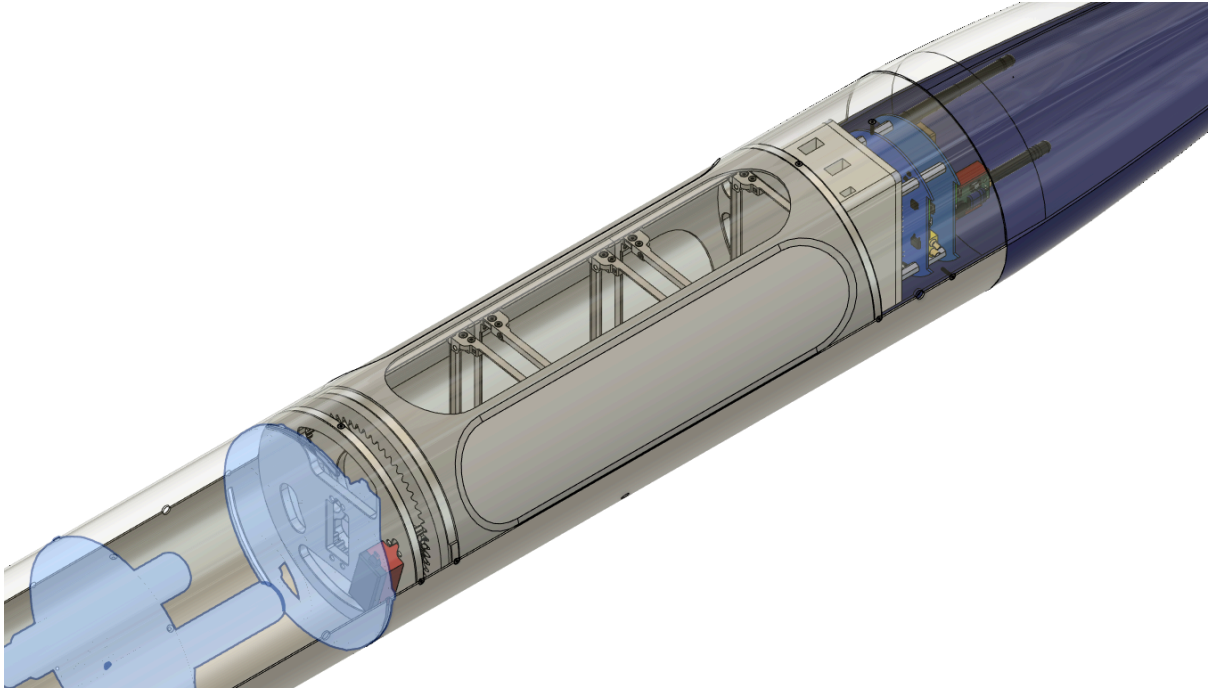


Figure 8: Payload location with the Air brake system

Below the payload exposure mechanism is the air brake assembly, which allows the rocket to slow down by increasing its drag, this system will allow us to hit the desired altitude more precisely. There are three slots in the body tube to allow the air brakes to deploy. Finally underneath the air brakes is the forward recovery bulkhead, it houses both raptor deployment systems as well as a forged eye bolt to ensure a safe recovery of the upper section.

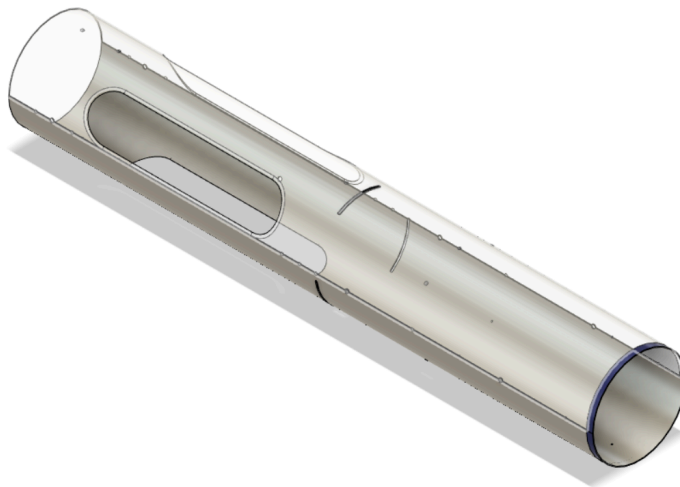


Figure 9: Upper body tube

C. Lower body tube

Regarding the lower body tube, it is 1005 mm long and it contains the back recovery components and the booster section. In the lower recovery bay section we can find the main and drogue parachutes, which will be deployed 5 seconds after apogee. The booster section contains the solid rocket engine, which will deliver the full thrust to the vehicle. A 12.725 mm thick aluminium thrust plate is placed on the top of the motor, it will also function as the aft recovery bulkhead and motor retainer. The recovery eye bolt is used to fasten the motor to the rust plate, which in turn adds to the structural integrity of the recovery system. This body tube also houses the rail buttons which will have an airfoil shape to reduce drag.



Figure 10: Lower body tube

D. Fin mount and fins

The four fins will be manufactured from 1/8" 6061-T6 aluminum plate due to its known mechanical properties and familiarity with the material. Stability and drag minimization were considered for the design of the fins, ensuring that the center of pressure of the rocket was located 2 calibers behind its center of mass. Fin thickness was driven primarily by the fin flutter velocity, the speed beyond which fin oscillation is amplified, potentially leading to catastrophic failure. This velocity is dependent on fin thickness and must be below the expected maximum speed. Two flutter boundary equations were used, the first corresponds to equation 18 in NACA Technical Paper 4197 and the second is a modification of the same equation that takes into account plate theory.

Using the maximum expected speed in case the air brakes fail to deploy and solving for fin thickness a minimum fin thickness of 1.844mm for equation 1 and 1.453 for equation 2 were found. A 1/16" (1.5875mm) plate could be used but would be too close to the fin flutter boundary, a 1/8" plate (3.175mm) would be much better suited and would withstand a speed of 2935 or 4151 ft/s, based on equations 1 and 2 respectively.

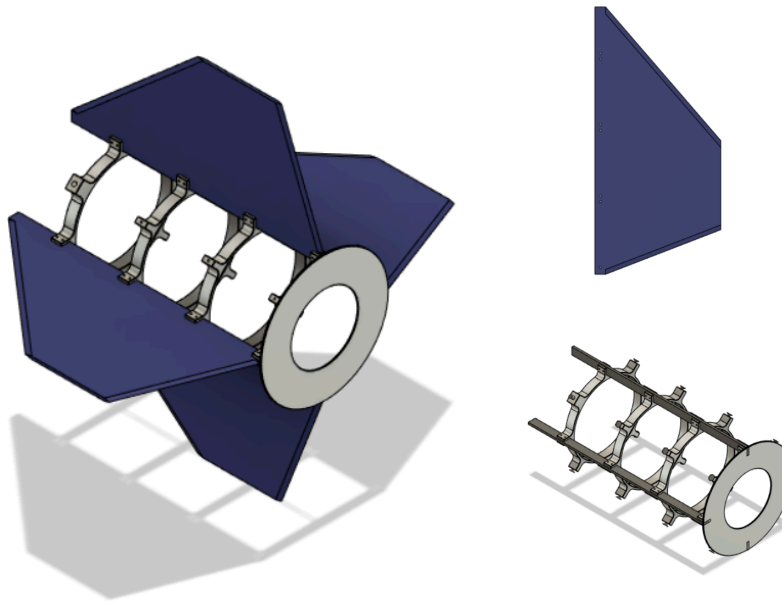


Figure 11: Fins and fin mount blueprints

The fins will be attached using four internal 6061 aluminum structures. These mounts serve both as motor centering rings and as fin mounts, additionally the forwardmost structure will have an attachment point for the lower rail button and the aft structure will be the rocket closure. The use of internal structures minimizes drag and allows for through the body fin mounting of metal fins. All 4 structures have a thickness of 0.5" or 12.7 and attach to the body tube using M2 flat head screws.

a. Side load / stiffness test

A static stress test was simulated using fusion 360's simulation environment. A force equivalent to the rocket mass (~241 N) acting upon the fin tip perpendicular to the fin root was simulated. The maximum lateral deflection must not exceed 10° and the minimum safety margin should be equal to or higher than 2.0.

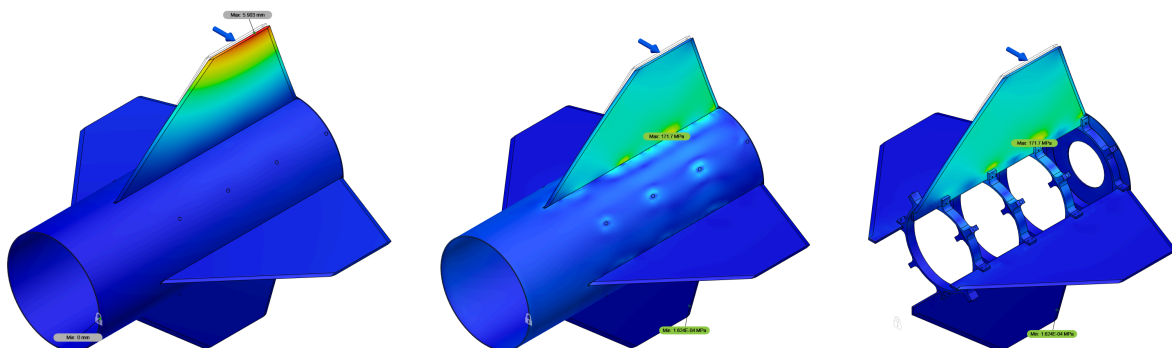


Figure 12: Side load and stiffness test results made with Fusion 360 static stress test.

The maximum lateral displacement was 5.9mm or 2.6° and the minimum safety factor was 2.95.

b. Fin longitudinal loading test

A static stress test was simulated using fusion 360's simulation environment. A force equivalent to two times the fin's mass times the rocket's maximum acceleration (~ 208 N) acting upon the fin tip parallel to the fin root was simulated. The minimum safety margin should be equal to or higher than 2.0.

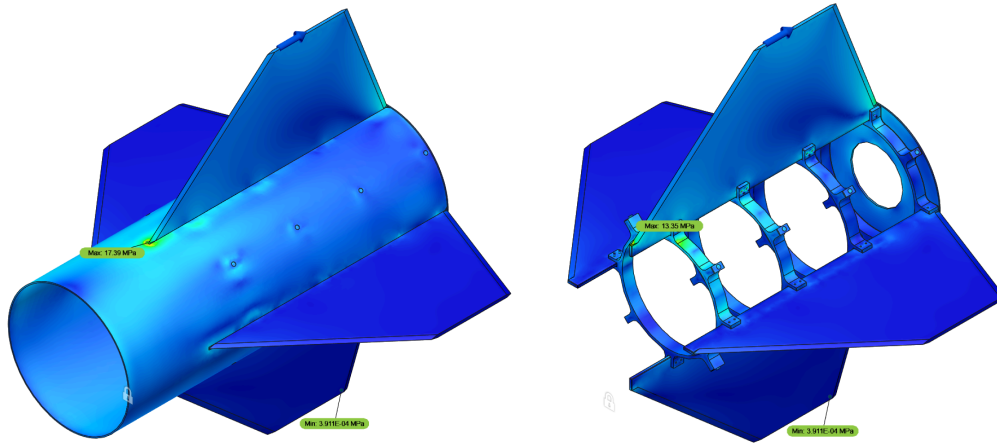
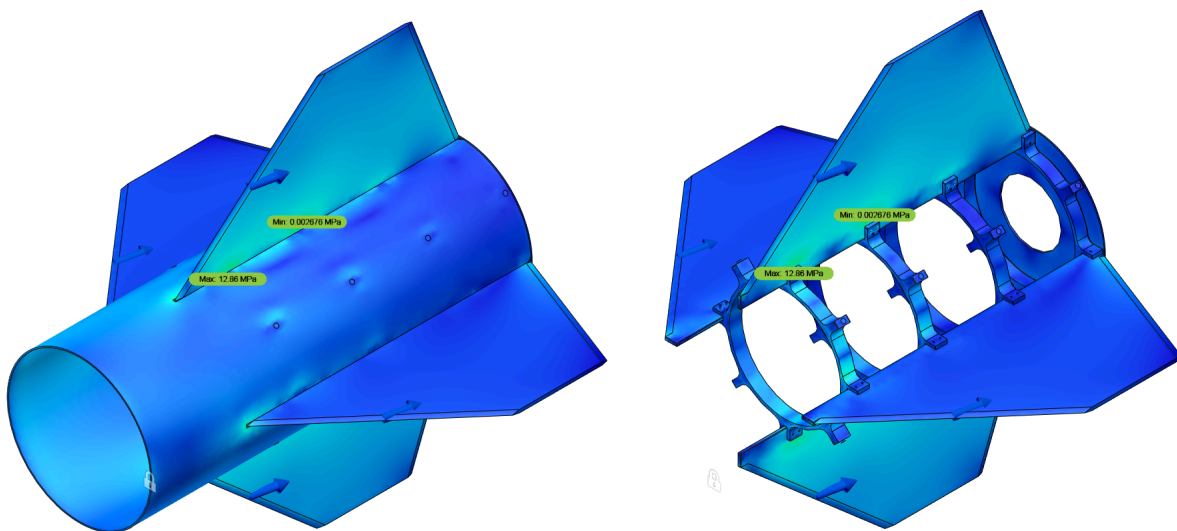


Figure 13: Fin longitudinal loading test results made with Fusion 360 static stress test.

The minimum safety factor was 9.799.

c. Drag force test

A static stress test was simulated using fusion 360's simulation environment. A force equivalent to the overall maximum drag force, including nose cone drag (~ 450 N) acting upon the fin's leading edge, parallel to the fin root, was simulated. The minimum safety margin should be equal to or higher than 2.0.



2. Avionics

The avionics system is based on two independent measurement systems, a Student Researched and Developed (SRAD) and one Commercial off-the-shelf system (COTS). The recovery system has four deployment charges: two CO₂ charges for rocket separation/drogue chute deployment and two Tender Descenders for main chute deployment. Each deployment charge will be wired to both the SRAD and COTS systems.

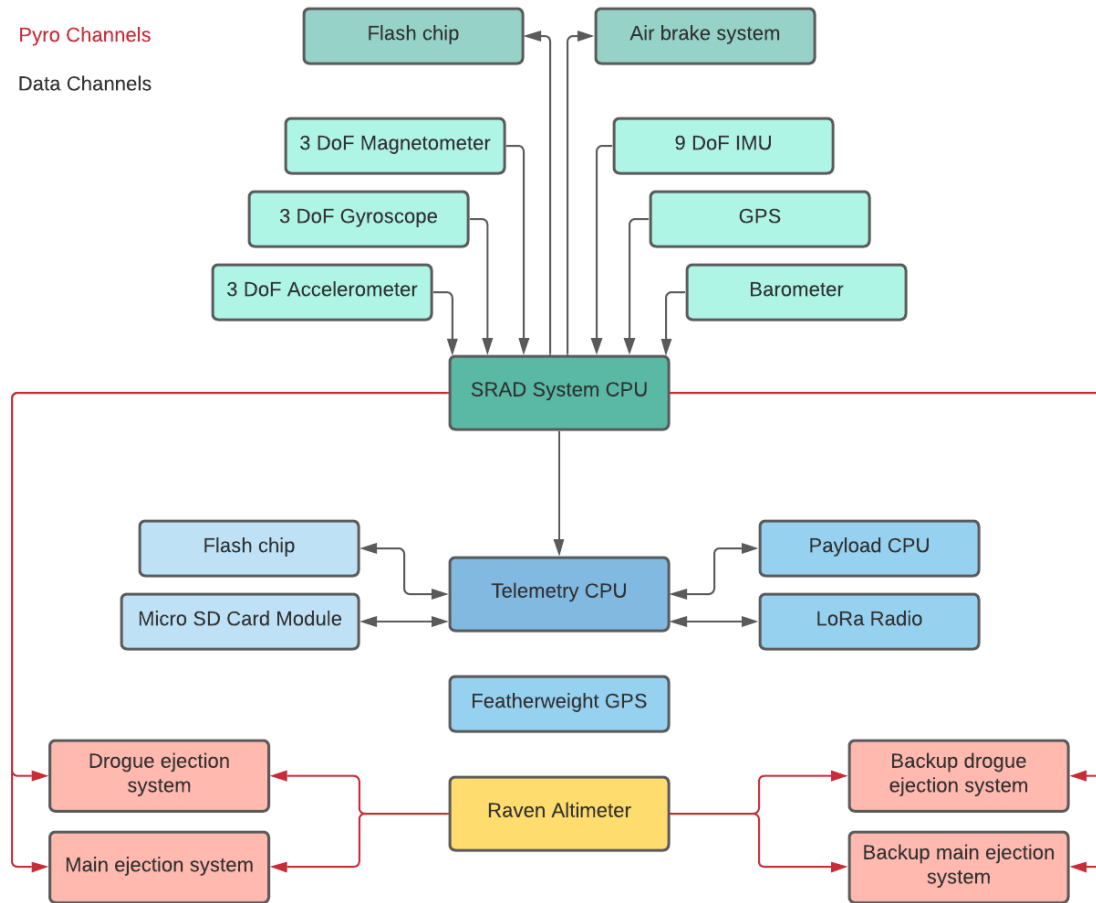


Figure 14: Avionics' general diagram

a. SRAD

Student Researched and Developed (SRAD) is a subsystem in charge of measuring air pressure, altitude, acceleration, and orientation. This information is crucial because it helps us to understand the behavior of the rocket's trajectory, predict the current and maximum altitude and fire recovery events. A development board (Teensy) controls the subsystem, processing the sensor data from two barometers, two 9 DoF inertial measurement units (IMUs), and a

GPS chip. It fuses it using a Kalman filter and runs a state machine to control recovery events and the rocket air brakes. It also sends the data to the telemetry system and a memory card. The SRAD subsystem has 4 pyro channels for the primary and backup charges used for main and drogue deployment.

b. COTS

Since our student group doesn't currently have the HAM Licencing, COTS instrumentation will consist of a Featherweight GPS in accordance with the IREC GPS Tracking Requirement and Recommendations and a Featherweight Raven altimeter. We will also use an SRAD GPS system using a 433Mhz LoRa module for ground recovery. All electrical components will have their independent batteries to ensure full redundancy.

c. Telemetry

Telemetry is a subsystem in charge of sending the data received from SRAD and COTS to the earth information station and storing the data in two different memory cards. The data will be downlinked using RFM95W LoRa Radio transceivers.

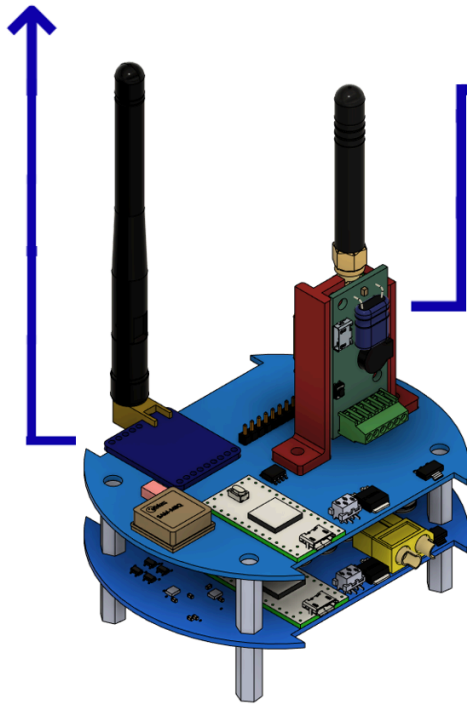
Table 3: Electronic components of the EFB avionics subsystem

SRAD	COTS	Telemetry
MCU PJRC Teensy 4.0 USB Development Board	Altimeter 1: Raven4 Altimeter	MCU: PJRC Teensy 4.0
9DOF IMU: Adafruit TDK InvenSense ICM-20948 9-DoF IMU	GPS: Featherweight GPS Tracker with battery and Ground Station.	Secondary Data Log: GD25Q16 - 2MB SPI;
Barometer: Adafruit BMP388		SD Card Log: Adafruit Micro SD SPI
9DOF IMU 2: Adafruit Precision NXP 9-DOF Breakout Board - FXOS8700 + FXAS21002		RF Radio: Adafruit RFM95W LoRa Radio
Barometer 2: LPS22 Pressure Sensor		Source of 3.3 V at 1.2 A
GPS: SAM-M8Q Chip Antenna		
Secondary Data Log: GD25Q16 - 2MB SPI		
Source of 3.3 V at 1.2 A		

The printed circuit board of SRAD, COTS, and Telemetry are in **appendix X**

Telemetry

- MCU: PJRC Teensy 4.0
- Secondary Data Log: GD25Q16 - 2MB SPI
- SD Card Log: Adafruit Micro SD SPI
- RF Radio: Adafruit RFM95W LoRa Radio



COTS

- Altimeter 1: Raven4 Altimeter
- GPS: Featherweight GPS Tracker with battery and Ground Station

SRAD

- MCU: PJRC Teensy 4.0 USB Development Board
- 9DOF IMU: Adafruit TDK InvenSense ICM-20948 9-DoF IMU
- Barometer: Adafruit BMP388
- 9DOF IMU 2: Adafruit Precision NXP 9-DOF Breakout Board - FXOS8700 + FXAS21002
- Barometer 2: LPS22 Pressure Sensor
- GPS: SAM-M8Q Chip Antenna
- Secondary Data Log: GD25Q16 - 2MB SPI

Figure 5: Avionics' components distribution

3. Propulsion

The selected propulsion system is a single student researched and developed (SRAD) solid rocket motor (SRM) using Potassium-Nitrate Coarse Sorbitol propellant (KNSB Coarse). Burn duration is 3.56s with an average thrust of 2546N and an maximum expected operating pressure of 6.87 MPa. Total impulse is 9203 Ns and specific impulse is predicted to be 134.2 s. The motor classifies as an M2550. The weight of the loaded motor is 12.65 kg.

The motor measures 907mm in length and 114.13mm in diameter. The propellant is divided into four Ballistic Test and Evaluation System (BATES) grains. Spacers are used to separate grains to ensure ignition and combustion of end faces.

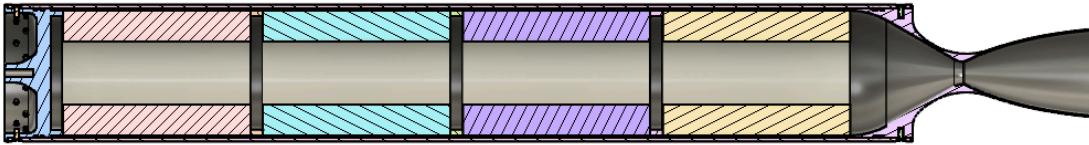


Figure 15: Propulsion System

a. Propellant

The chosen propellant is Coarse Potassium-Nitrate Sorbitol propellant (KNSB Coarse). Coarse refers to the KNO_3 particle size distribution (mean $411.4 \pm 30.7 \mu\text{m}$). Sorbitol particle size should be around $362.1 \mu\text{m} \pm 8.4 \mu\text{m}$. Casting tubes will be custom made from 10 point (0.254mm) paper. All in accordance to Olde, et al [1] and Gudnasson [2].

A propellant manufacturing protocol was developed based on existing literature and will be modified based on experience gained during characterization motors' manufacturing and testing. The propellant team also developed a propellant safety protocol, propellant sourcing, handling and storage protocol and design of the characterisation and flight motor casting equipment.

The propellant grains will be cast under pressure to prevent casting tube separation and air pockets that could lead to a rapid unscheduled disassembly of the motor

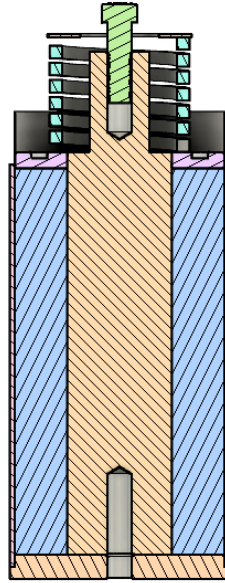


Figure 16: Propellant casting equipment

b. Case

The case will be made up from five main components: The motor casing, bulkhead, nozzle, seals and fasteners. The motor casing will be made from 6061-T6 cold formed, seamless aluminum tubing , the bulkhead and nozzle from AISI 1018 cold rolled steel, the seals will be size 155 high temperature Nitrile (Buna) o’rings and the bulkhead and nozzle will be fastened with 24 M4 high tensile steel flat head hex drive screws each to allow the motor to slide through centering rings. Case design was driven based on design and burst pressures for pressurized cylinders, bolt fastener theory and ASME standards for pressure vessel bulkhead design. Nozzle geometry is set to be a bell nozzle with an exit angle of 5° , corresponding to 108% length of a 15° conical nozzle and a theoretical nozzle efficiency of 99.6%.

To speed up the design process Richard A.Nakka’s Solid Rocket Motor Performance excel spreadsheet, used for internal solid rocket motor ballistics simulation was modified to include four additional spreadsheets used for casing and bulkhead design. They calculate the necessary case dimensions by using hoop stress, axial stress, tear out stress, bearing stress, shear stress along with an o’ring database to select suitable o’rings used for sealing the motor case.

Our bell nozzle profile was designed using Rocket Propulsion Analysis expecting a 99.6% efficiency which is related to the 5° exit angle. In terms of its manufacture it is made from Cold rolled AISI 1018 steel, and as designed in Fusion 360, it will be fixed to the combustion chamber using 24 M4x0.7 8mm bolts.

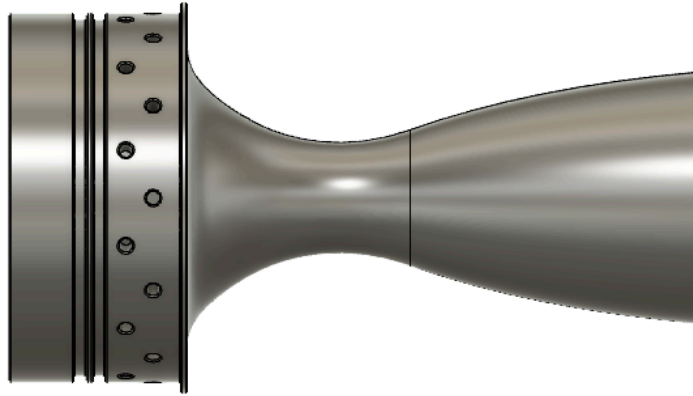


Figure 17: Nozzle

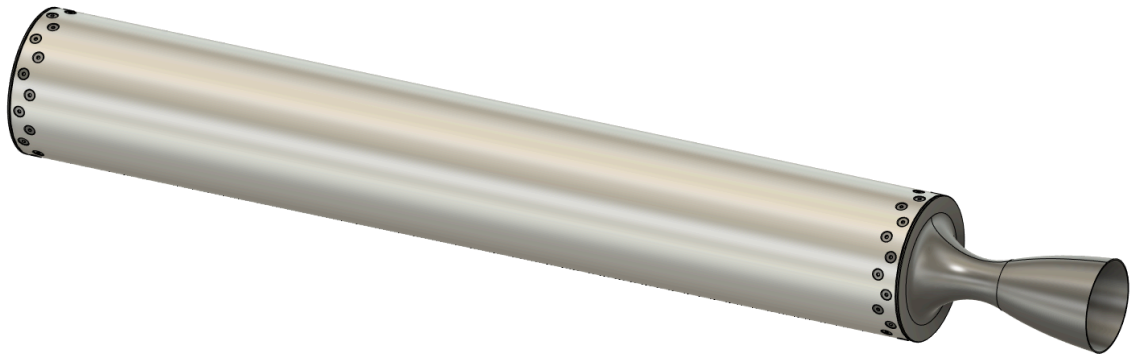


Figure 18: Propulsion system case

c. Adhesive

The currently planned adhesive for motor integration will be high-temperature RTV, also known as Red RTV, due to its excellent adhesive properties, temperature resistance and availability.

d. Thermal liner

The thermal liner will be constructed from 45mil EPDM rubber, as it is a commonly used material in the aerospace industry, requires minimal manufacturing and has excellent insulative properties.

The heat transfer analysis was done with THERMCAS, a thermal analysis package that determines the temperature distribution through the thickness of a motor casing wall that results from convective heating due to propellant combustion and resulting gas flow developed by Richard Nakka. Due to windows incompatibility issues it must be run on DOSBox, an x86 emulator with DOS.

e. Test Stand

A motor test stand was developed to fit motors from 1 to 5"OD up to 60"L, it is able to measure up to 9500N, casing temperature at 3 points and motor pressures up to 1600psi. The test stand structure is made from welded low carbon steel 1" caliber 10 square beams and 1/4" steel plates. The motors are held in place using U-bolts and 1"x 1/8" flat bars.



Figure 19: Test stand design

Simulation

The simulation team has been working with programs for motor internal ballistics simulation, thermal analysis, computational fluid dynamics and finite element analysis. For motor internal ballistics a couple of programs have been used, most notably the Nakka SRM excel sheet, which has been extensively modified and optimized; OpenMotor, a program designed by the MIT Rocket team and BurnSim, a commercial program conventionally used for IREC. For thermal analysis THERMCAS run on DOSBox is currently in use. Finite element analysis in both Fusion 360 and Solidworks; along with CFD in ANSYS Fluent.



Figure 20: Structural simulation of the test stand using Fusion 360 at 950 N of force

Table 4: Motor Output Data

Grain Mass (lb)	Total Impulse (lb/s)	Average Thrust (lb)	Thrust time (sec.)	Specific Impulse (sec)	Motor Classification
15.417	2069	518.4	3.559	134.2	M 2586

Table 5: Thrust Time Data

Data	Time (sec)	Thrust (N)	Thrust time (lb)
1	0.000	0	0
2	0.180	2753	619
3	0.184	2754	619
4	0.348	2787	627
5	0.516	2814	633
6	0.683	2834	637
7	0.850	2849	640
8	1.017	2857	642
9	1.184	2859	643
10	1.351	2855	642
11	1.518	2845	640
12	1.685	2828	636
13	1.852	2806	631
14	2.020	2777	624
15	2.188	2742	617
16	2.356	2702	607
17	2.525	2655	597
18	2.695	2602	585
19	2.866	2544	572
20	3.037	2480	557
21	3.209	2410	542
22	3.383	1436	323
23	3.563	0	0

4. Recovery

Recovery events are as outlined in the IREC Design, Test, & Evaluation Guide. A drogue chute will be released at apogee and the main chute will be released at 984.30 ft. The 3ft kevlar drogue chute will slow the descent rate to ~82 ft/s and the 10 ft kevlar main chute will reduce the descent rate to ~20.9 ft/s. Avionics are as described in the avionics section.

In our design the recovery subsystem separates the rocket in two halves. Each one will be connected by a shock cord to an eye bolt in each of the corresponding recovery bulkheads. The drogue chute will deploy and connected to it will be the two halves of the rocket and the main chute. This configuration was chosen due to the avionics bay being inside the RF-transparent nose cone.

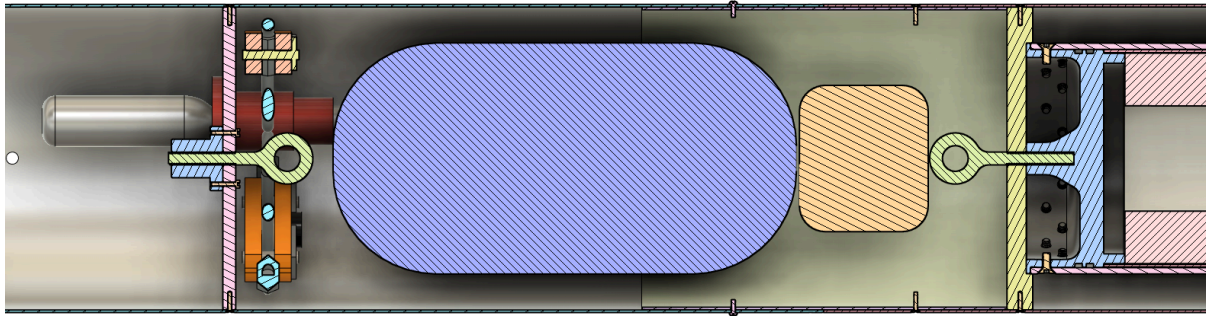


Figure 21:

The recovery system is composed of 4 deployment charges to ensure full redundancy. Two Tinder Rocketry Peregrine Raptors with 23g CO₂ cartridges for rocket separation/drogue chute deployment and two L2 Tender Descenders for main chute deployment. Figure X shows the schematic of the deployment system and Table X corresponds to the main characteristics of the recovery subsystem.

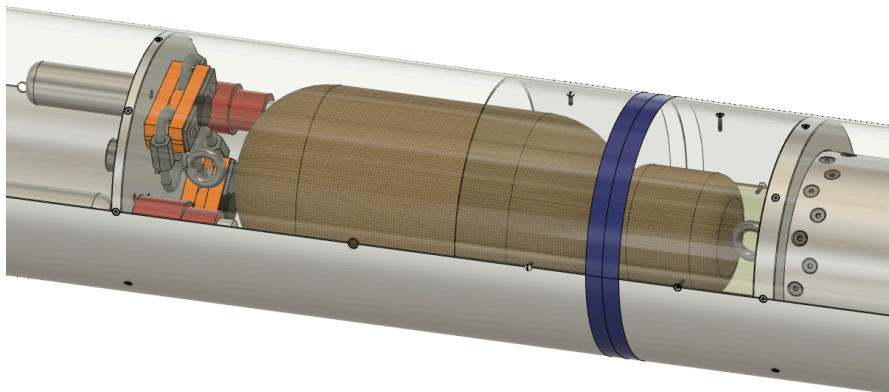
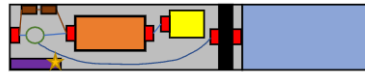
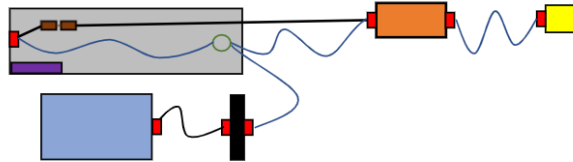


Figure 22: Schematic view of the recovery bay

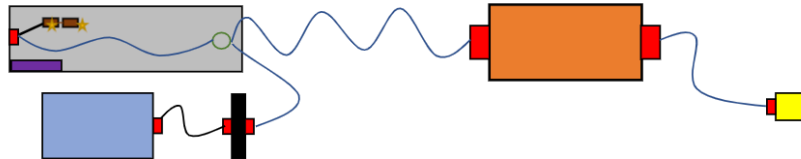
1. Raptor CO2 system pressurizes the recovery chamber. Breaking apart the shear pins and pushing the piston to separate the two halves.



2. The release of pressure separates both halves and deploys the drogue chute while the main chute release system holds back the main chute until the desired time.



3. The main chute mechanical release system opens and deploys the main chute at the desired height.



Color	Element
Orange	Main chute
Yellow	Drogue chute
Black	Piston head
Brown	Main chute mechanical release system
Purple	Raptor CO2 system
Red	Bolt rings
Green	Swivel ring
Grey	Lower body of the rocket
Blue	Upper body of the rocket

Figure 23: Description of the recovery subsystem function

A big challenge we had to solve was routing the pyro wiring from the nose cone all the way down to the recovery bulkhead. The wires must pass through the payload bay, payload exposure mechanism and air breaks to reach the recovery bulkhead. To allow this the air brakes, payload bulkhead and air brake bulkhead have slots that allow the wires to pass through without harm even during air brake operation.

VI. Flight simulations

Flight simulations were carried using the Open Rocket Software. Figure X shows the results of the vehicle without the deployment of the Air Brake System. Atmospheric conditions, launch site coordinates and wind conditions are shown in Figure 23 whereas simulation conditions are shown in Figure 23.

Further simulations including the Air brake system will be developed in the forthcoming weeks.

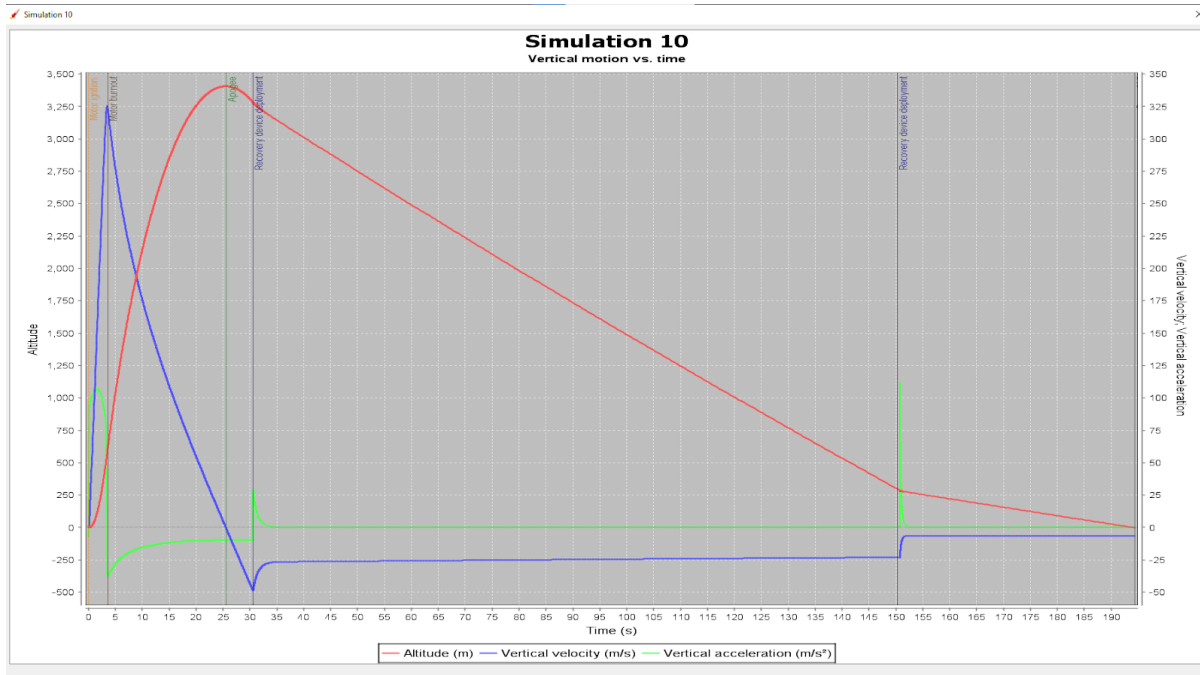


Figure 24: Flight profile of EFB obtained by Open Rocket. The red line corresponds to the altitude (m), the blue corresponds to the vertical velocity (m/s) and the green line represents the vertical acceleration of the vehicle (m/s²)

Mostrar la simulación

Nombre de la simulación:

Configuración del motor:

Condiciones del lanzamiento | Opciones de simulación

Viento

Velocidad media del viento: m/s

Desviación estándar: m/s

Intensidad de la turbulencia: % Medio

Wind direction: °

Lugar de lanzamiento

Latitud: ° N

Longitud: ° E

Altitud: m

Condiciones atmosféricas

☒ Usar los patrones atmosféricos internacionales.

Temperatura: °C

Presión: mbar

Varilla para lanzar

Longitud: cm

☒ Always launch directly up-wind or down-wind

Ángulo: °

Dirección: °

Wind direction, 0-360 degrees.
0 means from the north,
90 is from the east

Restaurar por defecto | Guardar por defecto

Simulación y Gráfico | Cerrar

Mostrar la simulación

Nombre de la simulación:

Configuración del motor:

Condiciones del lanzamiento | Opciones de simulación

Opciones del simulador

Método de cálculo: Barrowman Extendido

Método de simulación: 6-DOF Runge-Kutta 4

Cálculos geodésicos:

Duración de la etapa: s

Restaurar por defecto

Simulation extensions

Simulation extensions enable advanced features and custom functionality during flight simulations. You can for example do hardware-in-the-loop testing with them.

Add extension

No simulation extensions defined

Simulación y Gráfico | Cerrar

Figure 25: Atmospheric conditions and simulation options used for the flight analysis of the EFB

VII. Validation Tests

Although we could not construct our vehicle due to the COVID-19 Pandemic, the Rocket Laboratory has planned several tests in order to ensure a safe flight and landing. Table 6 gathers the planned tests for the aerostructures, avionics, propulsion and recovery. Payload validation tests are included in the avionics subsystem.

Table 6: Validation tests of the EFB Mission.

Test Number	Area	Name of the test	Description
1	Aerostructure	Wind Tunnel Aerodynamic	Scale model
2	Aerostructure	Axial Stress Load Test	
3	Aerostructure/Avionics	Airbrake System Deployment	
4	Propulsion	Propellant Characterization (Static Hot Fire 1)	Scale Motor
5	Propulsion	Propellant Characterization (Static Hot Fire 2)	Scale Motor
6	Propulsion	Thermal Ablative Characterization (Static Hot Fire 3)	Scale Motor
7	Propulsion	Scale Motor Test (Thermal, Manufacturing and Casting Protocol Validation)	Scale motor
8	Propulsion	Full Scale Motor Static Hot Fire Pressure test (Static Hot Fire 1)	Full Scale Motor
9	Propulsion	Full Scale Motor Static Hot Fire Thrust test (Static Hot Fire 2)	Full Scale Motor
10	Recovery	Drogue chute deployment test	
11	Recovery	Main chute deployment test	
12	Recovery/Avionics	Altitude-based deployment system	
13	Payload/Avionics	Data gathering	
14	All	Mission Simulation	

Appendix

APPENDIX A

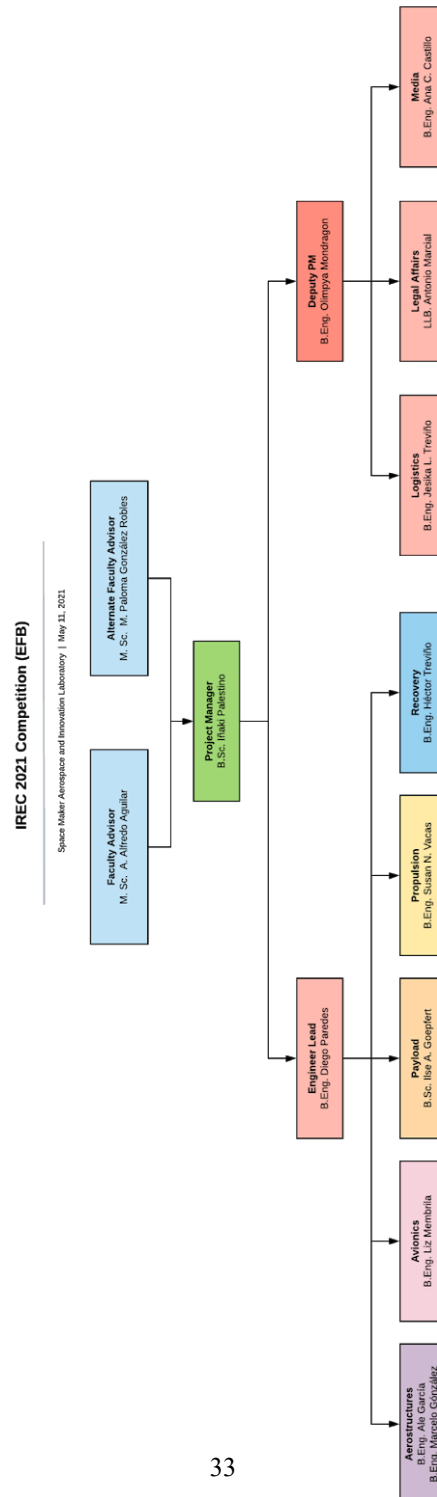
The following chemical reagents are used for the development of the EFB propellant. The used notation corresponds to the Global Harmonic System. More information regarding the precautionary statements during transportation, storage and disposal can be found at PubChem

Table 1:

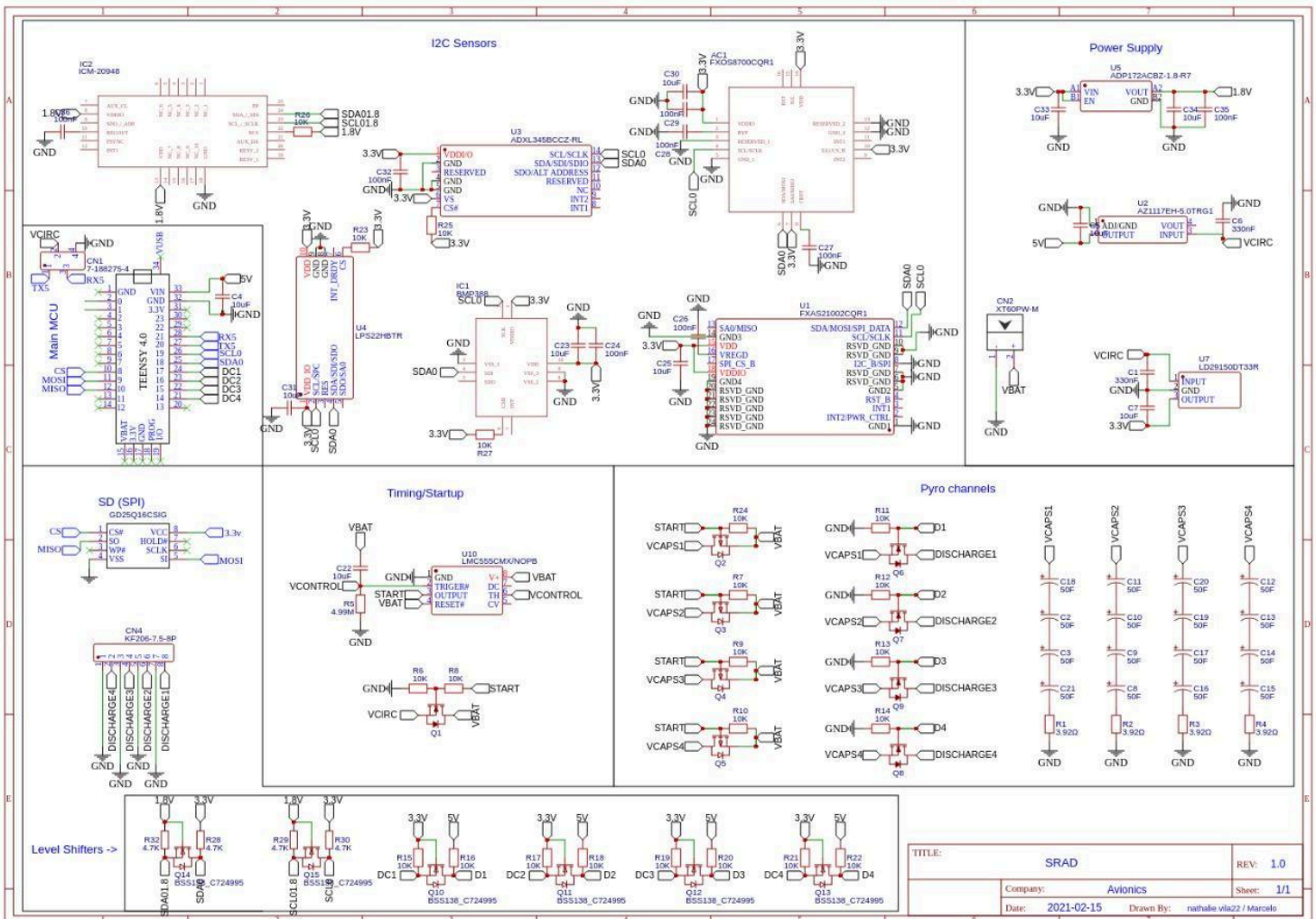
Reagent	Chemical Formula	LD ₅₀	Precautionary statements	First Aid	Pictogram
Potassium nitrate	KNO ₃	1901 mg/kg in rabbits	P210, P220, P221, P261, P264, P271, P280, P302+P352, P304+P340, P305+P351+P338, P312, P321, P332+P313, P337+P313, P362, P370+P378, P403+P233, P405, and P501	H272, H315, H319, H335	  Oxidizer Irritant
Sorbitol	C ₆ H ₁₄ O ₆	15900 mg/kg in rats			

APPENDIX B

EFB Team organigram



APPENDIX C



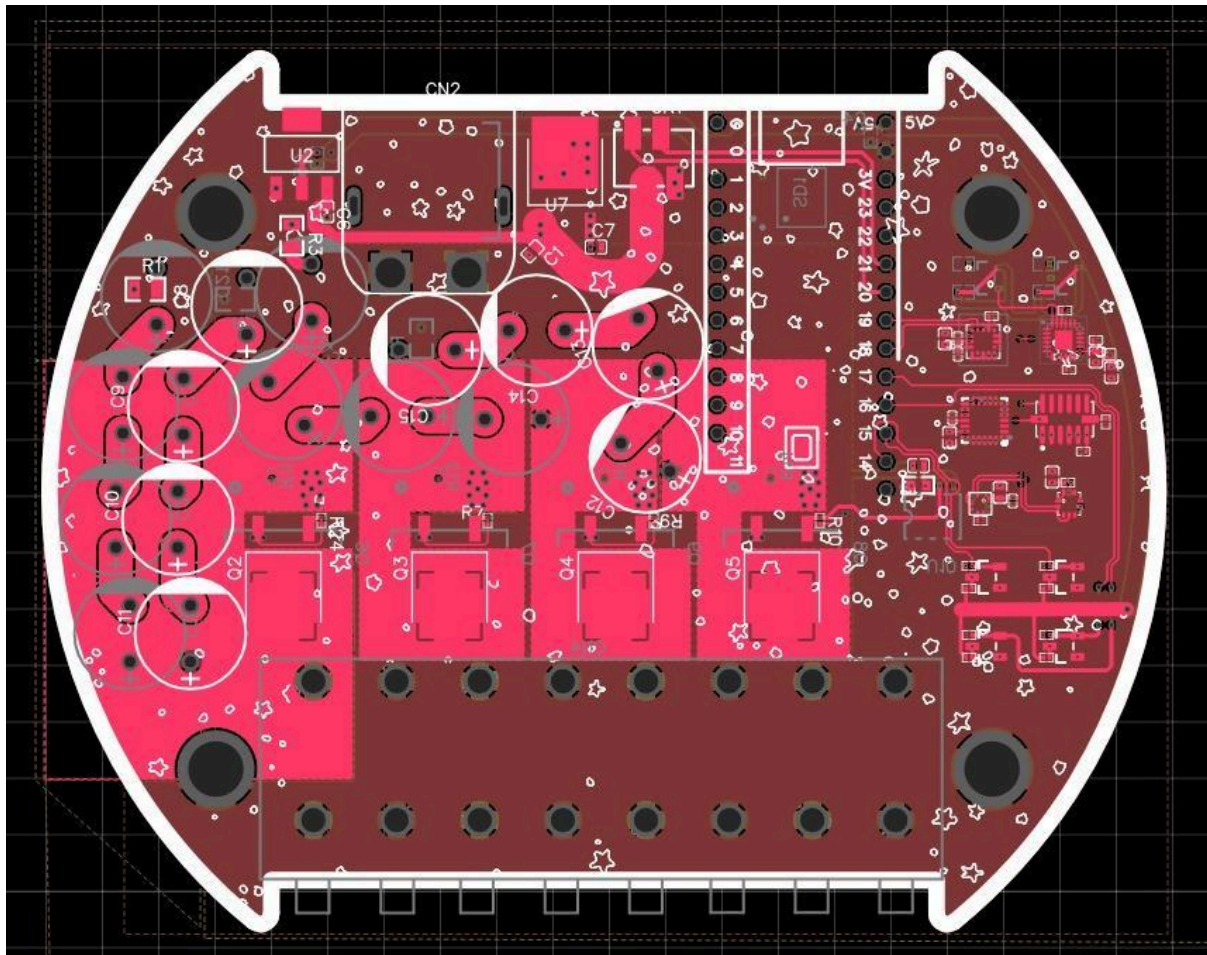


Figure X: Top Layer

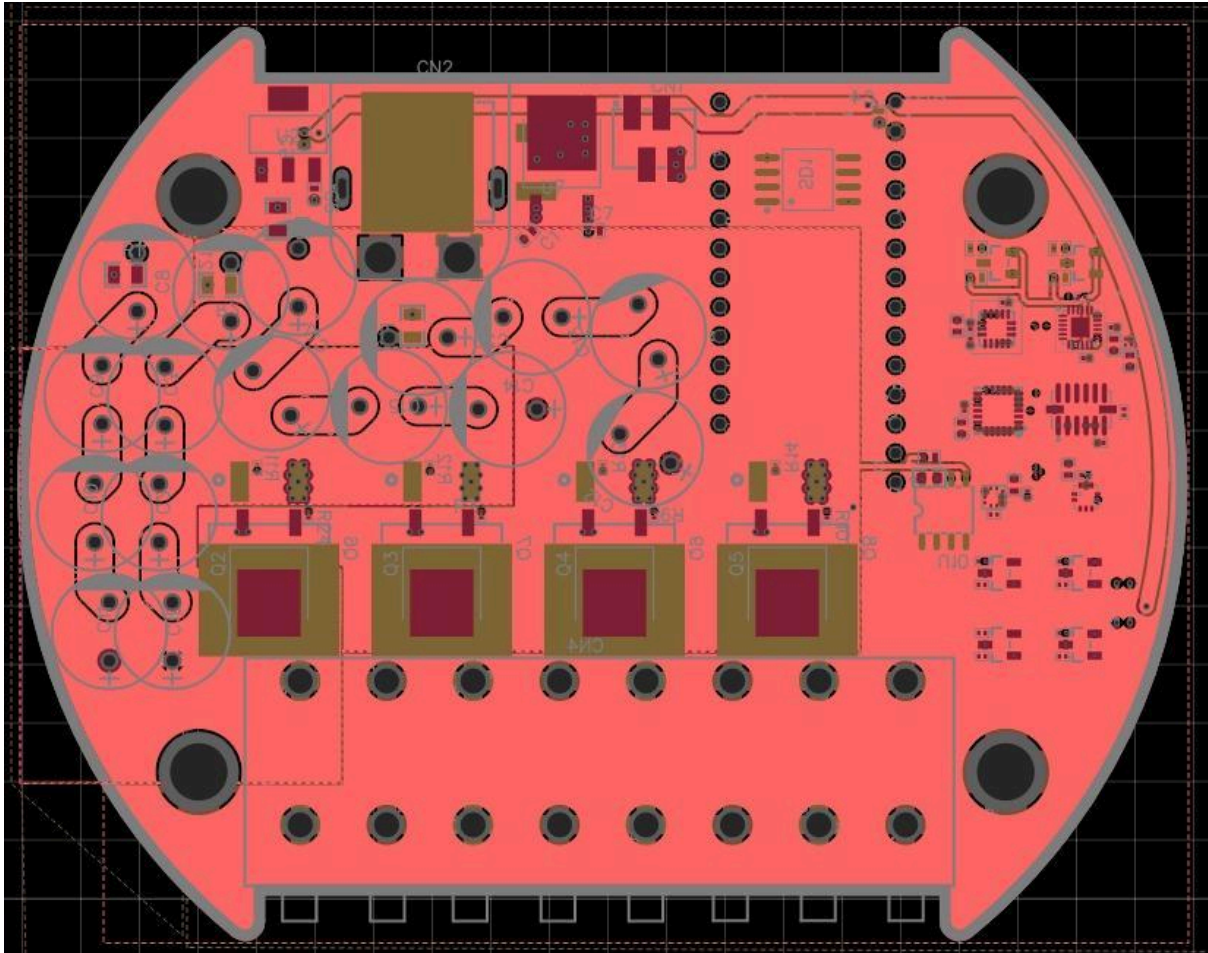


Figure X: Second Layer

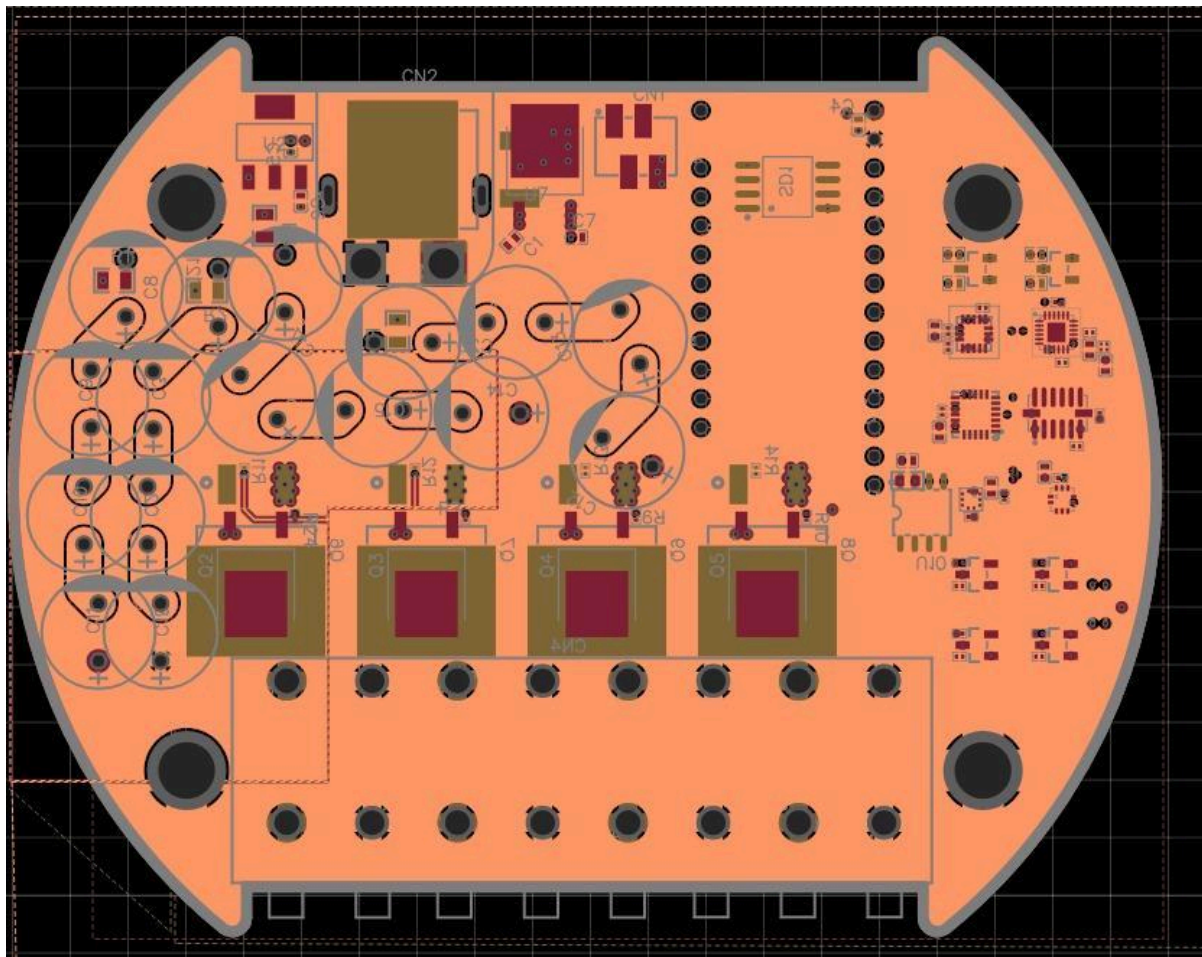


Figure X: Third Layer

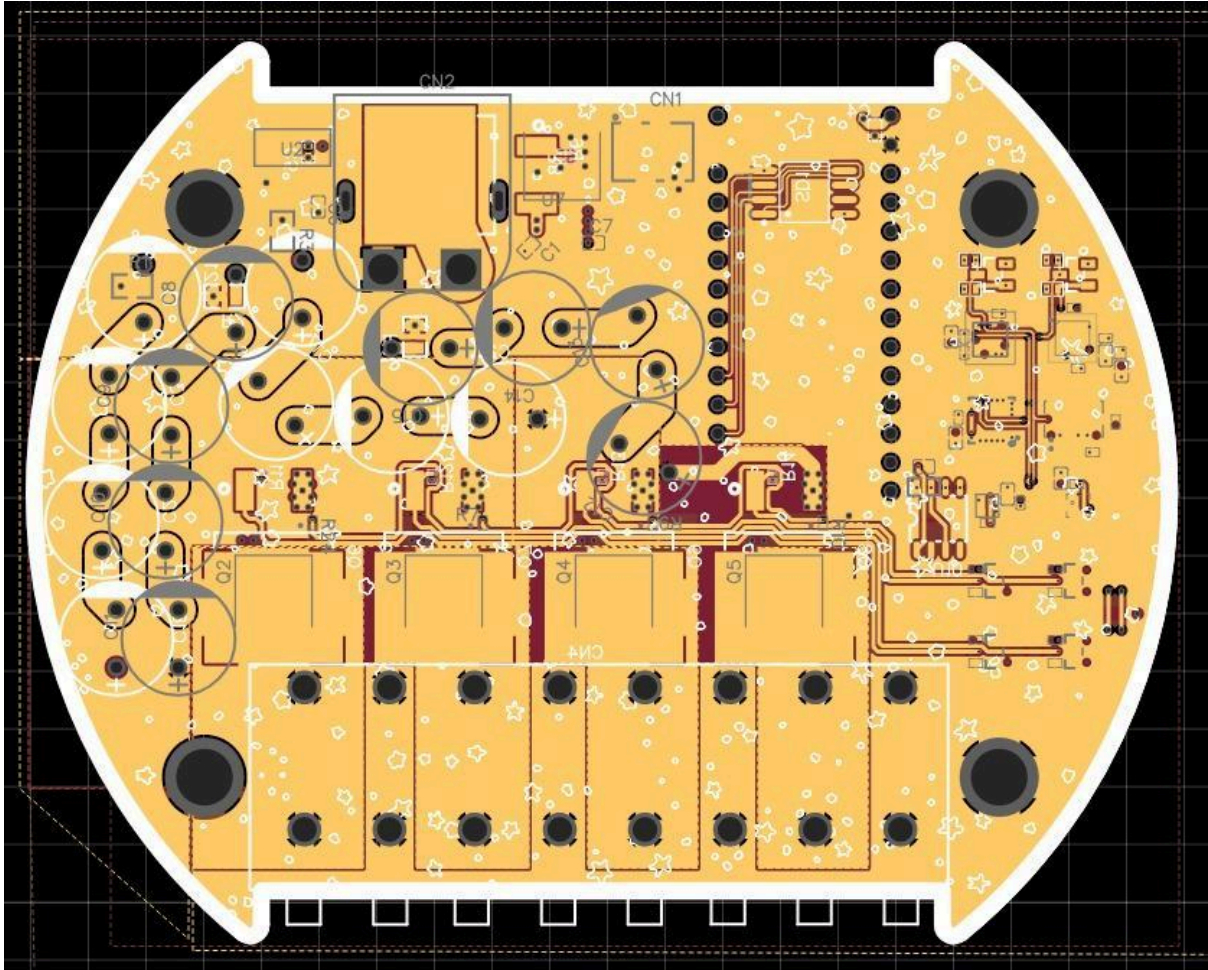


Figure X: Bottom Layer

APPENDIX D

Assembly protocol

Aft airframe

1. Centering ring
 - 1.1. Place rail button mount on centering ring slot
 - 1.2. Fasten using M2x0.4 6mm socket head screws
 - 1.3. Inspect centering ring for any damage
 - 1.4. Slide assembly into aft airframe and fasten using three M2x8mm flat head screws
2. Thrust plate
 - 2.1. Inspect for any damage
 - 2.2. Slide thrust plate into aft airframe and fasten using eight M2x8mm flat head screws
3. Motor
 - 3.1. Inspect motor for any damage
 - 3.2. Slide motor into the aft body tube, through the centering ring and into the thrust plate
 - 3.3. Fasten by installing recovery eye bolt through thrust plate and into the motor bulkhead
4. Fin assembly
 - 4.1. Slide all four fins into the fin mount retainer and fasten them using M2 8mm socket head screws (x4)
 - 4.2. Slide fins into mid fin mounts and forward fin mount, fasten them to the fins using M2 10mm socket head screws (x24)
 - 4.3. Inspect fin assembly for any damage
 - 4.4. Slide fin assembly into aft airframe and fasten using 11 M2x8mm flat head screws
5. Coupler
 - 5.1. Inspect for any damage
 - 5.2. Slide coupler into lower recovery bay and fasten using M2 8mm socket head screws (x4)
 - 5.3. Place round nuts over screws to protect parachutes and prevent tangling.
6. Rail buttons
 - 6.1. Instal rail buttons into centering ring and forward fin mount using two flat head 10-32 UNC x 1/2"screws

Forward airbreame

1. Recovery bulkhead
 - 1.1. Fasten eye bolt mount to recovery bulkhead using M2 12mm socket head screws (x4)
 - 1.2. Screw forged eye bolt through the recovery buckhead and eye bolt mount
 - 1.3. Assemble both Peregrine Raptor CO2 deployment devices
 - 1.3.1. Completely clean all components with a paper towel
 - 1.3.2. With a cotton tipped applicator use silicone grease to generously lube the Pyro Housing
 - 1.3.3. Use the cotton tipped applicator with silicone lube, to wipe a residue of lube on the inside surface of the charge cup
 - 1.3.4. Completely remove the plastic protective sheath from over the initiator heads and discard
 - 1.3.5. Cut a strip of blue painters masking tape about 1/2" x 3/4" and install on both initiators
 - 1.3.6. Use the cotton tipped applicator with silicon lube to wipe a residue of lube to the inside cavity of the charge cup. Be careful not to get any lube on the top
 - 1.3.7. surface of the charge cup!
 - 1.3.8. Pull both initiators to within an inch of the Dual Charge Cup.

- 1.3.9. Mix a small amount of 5 min epoxy. Dab a small amount of epoxy completely around the bottom of the initiators
- 1.3.10. Pull both initiators so that the tips do not extend past the top surface of the charge cup
- 1.3.11. Once the epoxy has hardened, measure out .2cc of either 3F (FFFG) Black Powder with the provided measuring scoop. Fill the provided scoop just to the line on the scoop
- 1.3.12. Pour powder into the prepared Charge Cup
- 1.3.13. Apply a Pyro charge cover disk or use a piece of 3M blue
- 1.3.14. masking tape over the charge cup to seal in the propellant. Trim the edges as necessary around the cup so the tape is flush with the edges
- 1.3.15. Carefully place the Charge Cup into the Pyro Housing
- 1.3.16. Carefully place the spring onto the Puncture Piston
- 1.3.17. Insert the Puncture Piston into the Pyro Housing
- 1.3.18. Attach Pyro Housing to the Mounting Cap
- 1.3.19. Instal 23g CO2 cartridge
- 1.4. Slide both raptors into the recovery bulkhead and fasten each one with two of the included mounting screws.
- 1.5. Thread igniter wires through the remain two mounting holes
- 1.6. Thread Tender Descender wires through remaining holes
- 1.7. Seal all holes with epoxy putty
- 1.8. Inspect all elements for any damage
- 1.9. Slide recovery bulkhead into upper body tube and fasten using four M2 8mm flat head screws
2. Air brake assembly
 - 2.1. Slide air brake spur gear into bottom air brake cover
 - 2.2. Place air brakes on top of bottom air brake cover and fasten using M4 8mm Socket Head Screws (x3). Make sure to align wiring leaf with the cover wiring slot
 - 2.3. Place top air brake cover over gears and instal slide guide screws M4 4mm Socket Head Screws (x3)
 - 2.4. Slide air brake mounts in line with their mounting holes and fasten using M2x0.4 8mm Socket Head Screws (x6)
 - 2.5. Inspect all elements for any damage
 - 2.6. Lube to ensure smooth rotation
 - 2.7. Fasten both LF20MG Metal gear servos to airbrake bulkhead
 - 2.8. Screw air brake spur gear (12 tooth) to central servo
 - 2.9. Screw 10 tooth spur gear to payload exposure mechanism servo
 - 2.10. Inspect all elements for any damage
 - 2.11. Thread wiring through air breaks and bulkhead
 - 2.12. Slide air brake bulkhead into the upper body tube and fasten using four M2 8mm flat head screws
 - 2.13. Fasten air brake assembly using three M2 8mm flat head screws
3. Payload exposure mechanism assembly
 - 3.1. Slide payload cover over payload bulkhead and internal spur gear
 - 3.2. Fasten to internal gear with four M2 8mm flat head screws
 - 3.3. Lube to ensure smooth rotation
 - 3.4. Inspect all elements for any damage
 - 3.5. Slide assembly into upper body tube and fasten using four M2 8mm flat head screws.
 - 3.6. Ensure spur gears fit together and have smooth rotation
 - 3.7. Thread recovery and servo wiring through payload bulkhead
4. Payload
 - 4.1. Ensure all payload systems are nominal

- 4.2. Connect payload exposure mechanism to payload
 - 4.3. Vertically integrate the payload
 - 5. Avionics
 - 5.1. Assemble avionics stack (stand off posts, telemetry/cots and SRAD board)
 - 5.2. Screw battery box to avionics bulkhead
 - 5.3. Place all four batteries in box, close and screw battery box
 - 5.4. Screw avionics stack to battery box
 - 5.5. Thread wires through avionic bulkhead
 - 5.6. Wire avionics (pyro channels, air brakes, batteries, payload)
 - 5.7. Final visual inspection
 - 5.8. Slide avionics bulkhead into the upper body tube and fasten using four M2 8mm flat head screws.
 - 6. Nose cone
 - 6.1. Inspect nose cone for any damage
 - 6.2. Slide nose cove over avionics stack and fasten using four M2 8mm flat head screws.
 - 7. Systems check
 - 7.1. SRAD instrumentation check
 - 7.2. Payload instrumentation check
 - 7.3. Air Brakes check
 - 7.4. Payload exposure mechanism check
- Integration
- 1. Pack main and drogue chutes into deployment bags
 - 2. Link and fold all shock cords
 - 3. Link Tender Descenders in series and attach to main deployment bag
 - 4. Slide main deployment bag into upper recovery bay
 - 5. Slide drogue deployment bag into lower recovery bay
 - 6. Slide upper body tube over coupler
 - 7. Fasten with four 2-56 X 1/4" nylon shear pins
 - 8. Inspect rocket for any damage